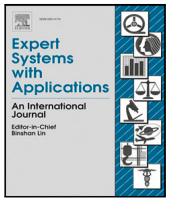
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Open-source multi-objective optimization software for menu planning

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ABSTRACT

Healthy nutrition is essential for maintaining health and preventing chronic diseases. However, with the increase in dining out, the prevalence of obesity and related chronic diseases is increasing daily. Therefore, to provide adequate and balanced nutrition in public, it is necessary to serve healthy menus following the energy and nutrient requirements of the target group in food service institutions. However, menu planning in food services is a complex process involving several factors. The planned menus should provide sufficient nutrients for the target group and consider factors such as color, consistency, appearance, and a variety of food groups. In addition, existing studies have limited constraints, and the fact that they are not open-source makes it difficult to conduct more comprehensive new studies. Furthermore, the lack of dietitian opinions limits its applicability in practice. Therefore, this study applied four different multi-objective optimization algorithms (AGEMOEA, SMSEMOA, NSGA2, and NSGA3) to solve the menu planning problems. The results were shared with 20 food service dietitians who are experts in the area and were asked to score the menus. In addition, an open-source tool called EvoMeal has been developed. In conclusion, our primary findings showed that AGEMOEA and SMSEMOA were used for the first time in the literature for the menu planning problem, which is a multi-objective problem, and they performed better than both NSGA2 and NSGA3 in general. In addition, expert opinions confirmed that AGEMOEA and SMSEMOA give better results than other algorithms. However, future studies need to conduct a more comprehensive expert evaluation to increase its applicability to the field.

1. Introduction

Adequate and balanced nutrition is essential in preventing chronic disease and maintaining health (Haddad et al., 2016). However, the data show that overweight and obesity prevalence has increased over the years globally, with the contribution of unhealthy nutrition (World Health Organization, 2013, 2022). Therefore, to manage and prevent obesity and related chronic diseases, individual diet plans have been developed, medical nutrition treatments have been suggested, and the number of studies in this area has been increasing recently (Canuto, Garcez, de Souza, Kac, & Olinto, 2021; Castro-Barquero, Ruiz-León, Sierra-Pérez, Estruch, & Casas, 2020; Magkos, Hjorth, & Astrup, 2020). However, individual interventions alone are insufficient for managing and preventing obesity. Furthermore, in recent years, people have been consuming too many meals outside the home, and over the years, the number of people who eat outside the home has been increasing (Mistretta, Caputo, Cellura, & Cusenza, 2019). Therefore, to provide adequate and balanced nutrition in public, it is necessary to serve healthy menus following the energy and nutrient requirements of the target group in food service institutions (Turkey Minstry of Health, 2020).

Studies have shown a positive association between eating outside the home and being overweight or obese (Bezerra, Curioni, & Sichieri, 2012; Bezerra, Moreira, Cavalcante, Souza, & Sichieri, 2017; Lima, de Almeida, Botelho, Nakano, et al., 2023). To prevent this, onsite and retail food service operations such as schools, kindergartens, nursing homes, and hospitals should focus on serving healthy menus to their target populations. Given the increasing prevalence of obesity and related chronic diseases, the role of food service managers and workers in providing adequate and balanced menus is becoming increasingly important (Mathews, Patten, & Stokes, 2021).

Menu planning is a complex problem involving many factors. The planned menus are expected to meet the target group's energy and nutrients, be sufficient in terms of quality and quantity, be compatible in terms of color, consistency, and appearance, and contain a wide variety of food groups in sufficient quantities. Therefore, although it contains various challenges, the menu is a valuable tool that plays a role in healthy choices, healthy eating habits, and nutrition education (Lima et al., 2023). A well-trained professional can handle the complexity of regular menu planning. However, considering all the abovementioned

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factors, the menu planning process can be complicated. Also, if a menu is planned for a special group, success may decrease as many different factors will come into play (Seljak, 2009). Therefore, menu planning using an artificial intelligence approach that considers all these factors could increase success.

Various algorithms have been used in the literature to solve the menu planning problem (Hernandez-Ocana, Chavez-Bosquez, Hernandez-Torruco, Canul-Reich, and Pozos-Parra, 2018; Marrero, Segredo, León, & Segura, 2020; Moreira, Wanner, Martins, & Sarubbi, 2017; Seljak, 2006, 2009). However, as far as we know, no open-source application has been made available for conducting repeatable experiments in the field of automated meal generation. This lack of access has greatly limited studies in this area. Our primary aim in this study is to address this gap by developing Evolutionary Automated Meal Generation (EvoMeal), an open-source software. Our application will allow for effortless integration of other algorithms available in the literature and pave the way for future studies to be conducted with ease. EvoMeal will enable the generation of personalized menus designed specifically to cater to various medical conditions. Besides, the Adaptive Geometry Estimation-Based MOEA (AGEMOEA) and the Multiobjective Selection-Based On Dominated Hypervolume (SMSEMOA) algorithms have yet to be used for this purpose. Therefore, another aim of the study is to solve the problem of menu planning using AGEMOEA and SMSEMOA. For the purpose of comparison of these algorithms, the Non-dominated Sorting Genetic Algorithm (NSGA2) and the Reference-point-based Many Objective Evolutionary Algorithm (NSGA3) multi-objective optimization algorithms are also implemented in our tool. As a last part of our study, we evaluated the menus' adaptability. We identified the best and worst menus based on the fitness value averages of non-dominant solutions. To assess their practicality in real life, we asked dietitians to evaluate them without telling them that they were created using a tool.

This article's contributions to both research and engineering can be summarized as follows:

- To develop new algorithms and ensure that others can easily replicate our results, compare tools, and reuse our algorithm implementations, we have made our tool prototype available under the open-source LGPL license. This can be accessed through the public hosting repository GitHub.[1]
- We used the AGEMOEA and SMSEMOA algorithms for the first time in the menu planning problem.
- We designed new fitness functions that can solve the menu planning problem.
- We created a new dataset consisting of a total of 332 dishes of various types, and with this dataset, we performed a 5-day menu planning.
- We evaluated the menus generated by algorithms through dietitians and discussed their applicability to the field.

The study experiments were divided into two parts. Firstly, the NSGA2, NSGA3, SMSEMOA, and AGEMOEA algorithms were compared and statistically tested. Secondly, dietitians evaluated the best and worst menus created by these algorithms according to menu planning principles to examine their practicality. The best and worst menus were chosen according to the average fitness values. This article was structured as follows: Section 2 outlined the materials and methods, Section 3 presented the experimental studies, and Section 4 concluded with a discussion and conclusion.

[1] https://github.com/omursahin/EvoMeal.

2. Material and methods

2.1. Principles of menu planning

The menu is a list of dishes served in food services. In addition, the menu is the most effective management tool in the food services organization. A well-planned menu also impacts other functions of the food service organization, such as purchasing, storage, and production. However, the menu planner tries to strike a balance between both consumers and food service. On the one hand, menus must comply with consumers' desires, eating habits, and cultures. On the other hand, the budget for the food service, the number of personnel, and the kitchen and dining hall features should be considered. While providing all of them, the menu should also contribute to adequate and balanced nutrition and meet consumers' energy and nutritional requirements (Theis, Payne-Palacio, et al., 2011). Menus include various food groups such as soups, main courses, vegetable dishes/side dishes, salads/garnishes, desserts, and fruits. A suitable menu is created by bringing these food groups together according to various rules.

Different cultures follow various criteria when planning menus. However, we have summarized the menu planning principles used in this study for this study below (Theis et al., 2011):

1. **Nutrient**: Meeting the energy and nutritional requirements of the target group: The primary aim of menu planning is to meet the energy and nutritional requirements of the target group. Therefore, age, gender, physical activity, and special physiological conditions (e.g., disease) are considered. The recommendations of national dietary guidelines are generally used to determine energy and nutrient requirements.
2. **Color**: It is not enough for menus to only meet the requirements. At the same time, eye-appealing dishes should be served. Menus presented in different colors are more attractive to the consumer. Therefore, it is not preferred that all dishes on a menu be the same color. For example, a menu consisting of mushroom soup, boiled chicken, and yogurt is entirely white and does not look pleasing to the eye. Serving tomato soup, boiled chicken, and salad would be more appealing to consumers.
3. **Texture/consistency**: Serving very dry or completely liquid meals together is not appropriate. There should be a balance in menus regarding the texture/consistency of food for the comfort of consumption.
4. **Repetition**: Constantly serving similar dishes on menus reduces consumer satisfaction and negatively affects the organization's success. Therefore, different dishes should be preferred on menus, especially on consecutive days.
5. **Variety**: To plan an adequate and balanced menu, one must ensure that the foods served are diverse. Variety can be achieved in meals by considering the food groups (such as meat dishes, vegetable dishes, and legume dishes) or the main components of the dishes (such as the dairy group, vegetable group, and legume group).

2.2. Problem definition

In the EvoMeal, each solution was created for n days, and each day consisted of 3 dishes: main dish, side dish, and fruit/dessert/salad or dairy products. The menu for a total of n days corresponded to a single solution, and there were as many menu solutions as the number of populations in a population. The specified representation was given in Fig. 1. Each dish was defined by IDs specified in integer type.

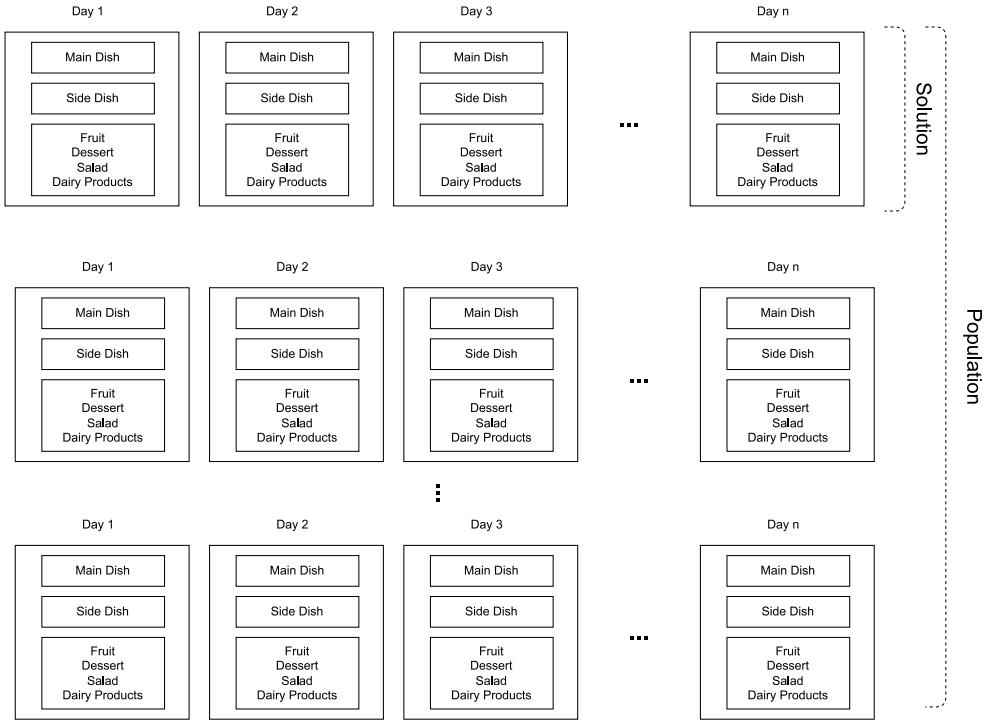


Fig. 1. The representation for menu planning problem.

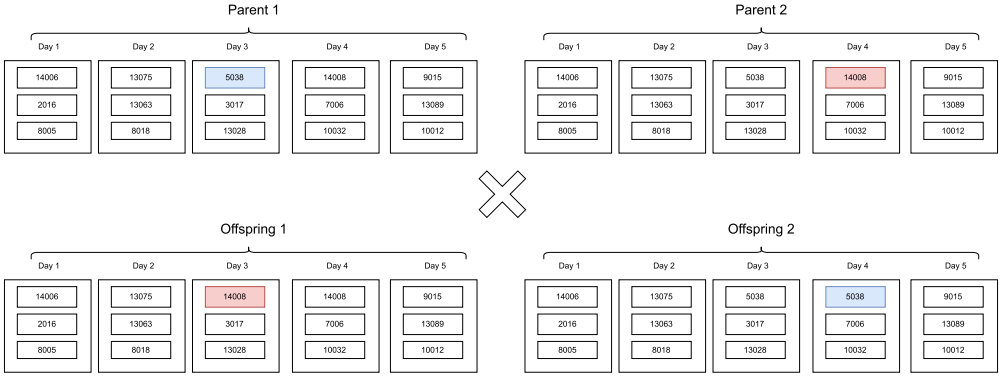


Fig. 2. Crossover operator.

2.3. Search operators

Two different search operators, crossover and mutation, were used to generate new solutions.

The crossover method given in Fig. 2 was used for data exchange between individuals. In the crossover performed dish in this study, firstly, in which dish type the crossover will be performed was determined. Then, the days between which data will be exchanged are determined, and these two dishes were swapped.

An example of a mutation operator was given in Fig. 3. In mutation, dish type and day were determined randomly and replaced with a new dish from the database.

2.4. Dataset collection

First, standard recipes for Turkish Cuisine dishes were found using various books and guides (Merdol, 2003; Turkey Minstry of Health, 2020). "Standard recipes" are the food recipes that have been tested, adjusted, and tested again for use by a specific food service establishment. They are designed to produce consistent results and yield each time, given that the exact procedures are used with the same type of equipment and the same quantity and quality of ingredients (US Department of Agriculture, 2002). Turkish Cuisine dishes whose standard recipes the authors could access were included in the study to calculate the meals' energy and nutritional values accurately. Then, the energy, carbohydrate, fat, and protein amounts in one portion of these dishes were calculated using the Nutrition Information Systems (BEBIS, Version 8.0) software. This software is the most widely used tool in calculating the energy and nutrient contents of foods in Türkiye. It combines the German Nutrient database, the United States Department of Agriculture (USDA) database, and the Turkish TurKomp database. Afterward, the dishes were classified according to the food types in Turkish Cuisine and based on the Menu Planning Guidelines (Turkey Minstry of Health, 2020), and a total of 19 food types were created

Table 1
The dish types in the dataset.

Code	Dish type	#	Code	Dish type	#
1	Seafoods	9	11	Patty and ravioli	20
2	Large piece of meats	15	12	Pilaw and pasta	29
3	Poultries	13	13	Egg dishes	2
4	Vegetable with meats	24	14	Salads	30
5	Legumes with meats	4	15	Desserts	51
6	Stuffed dishes	12	16	Fruits	15
7	Vegetables with olive oil	20	17	Small piece of meats	15
8	Legumes with olive oil	4	18	Meatballs	20
9	Stuffed vegetables with olive oil	5	19	Yoghurt, cucumber-yoghurt, ayran	3
10	Soups	41			

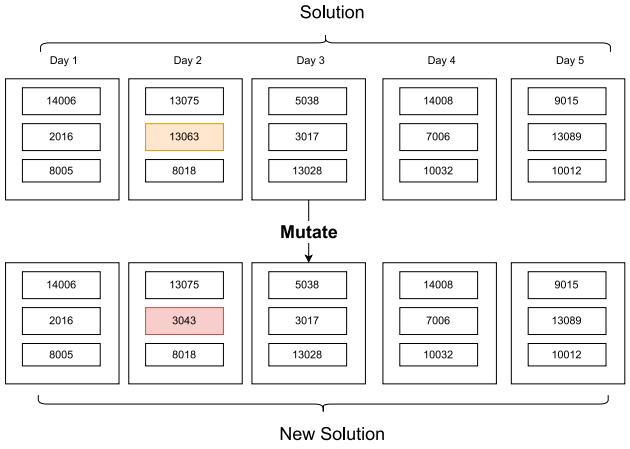


Fig. 3. Mutation operator.

(Table 1). The dataset consisted of a total of 332 different dishes, 116 of which were main dishes, 115 were side dishes, and 101 were fruit, dessert, salad, and dairy products. The maximum number of menu combinations for a day was $116 * 115 * 101 = 1\,347\,340 (\sim 10^6)$. The number of possible combinations for five days was $(10^6)^5$. The complete dataset is available as supplementary material.

2.5. Optimization algorithms

To solve the problem of planning a menu, six different criteria were defined, each of which conflicts with the others. While considering the nutritional content of the food (***nutrient***), an attempt was made to avoid repetition of food within meals (***repetition***) and not include food from the same meal group on two consecutive days (***meal group***). Additionally, a diverse and balanced menu was created by using a variety of main ingredients (***main ingredients***), and physical diversity was achieved through the use of color (***color***) and texture (***consistency***). Therefore, this study opted to utilize multi-objective optimization algorithms to achieve its objectives. In this paragraph, the fitness function names used in the study are given in ***bold and italic***.

In the EvoMeal, four distinct multi-objective optimization algorithms were utilized, namely Non-dominated Sorting Genetic Algorithm (NSGA2), Reference-point-based Many Objective Evolutionary Algorithm (NSGA3), Multiobjective Selection Based On Dominated Hypervolume (SMSEMOA), and Adaptive Geometry Estimation-Based MOEA (AGEMOEA). The study's algorithms were implemented using Multi-objective Optimization in Python (Pymoo)[2]. A brief description of the algorithms was given in the following subsections.

[2] https://www.pymoo.org.

2.5.1. NSGA2

The NSGA2 algorithm is a widely used algorithm that helps solve multi-objective optimization problems. It is an extension of the original NSGA algorithm and was introduced by Deb et al. in 2000 (Deb, Agrawal, Pratap, & Meyarivan, 2000; Deb, Pratap, Agarwal, & Meyarivan, 2002). It was developed to address the shortcomings of the original NSGA algorithm created by Srinivas and Deb (1994). NSGA2 is known for its robustness and efficiency in handling problems with multiple conflicting objectives. Due to its capabilities, it has become a popular tool in various fields such as engineering, finance, and data science.

NSGA2 is a search algorithm that determines the quality of solutions in a search space based on Pareto dominance (Kuo, Edbert, Zulvia, & Lu, 2023). A solution dominates another if it is at least as good in all objectives and strictly better in at least one. This helps NSGA2 to identify a set of solutions that are not dominated by any other, forming the Pareto front. The Pareto front represents the trade-off between conflicting objectives and is the set of solutions that are considered optimal. NSGA2 uses a metric called crowding distance to maintain diversity in the population. This metric is used to measure how crowded a solution is in the objective space. Solutions that are located far from the densely populated regions of the objective space are given preference to ensure that a diverse set of non-dominated solutions is preserved. NSGA2 has an efficient non-dominated sorting mechanism that is one of its key features. This mechanism organizes the solutions into different fronts based on their Pareto dominance. The best solutions are assigned to the first front, while the lower quality ones are assigned to subsequent fronts. This ensures that the best solutions are always identified. The brief steps of NSGA were given in Alg. 1.

Algorithm 1 NSGA2 Algorithm

```
Initialization of the population
Evaluation
repeat
    Fast non-dominated sorting
    Crowding distance assignment
    Selection and crossover
    Mutation
    Survival selection
until A termination criteria is satisfied
```

2.5.2. NSGA3

The NSGA3 algorithm (Deb & Jain, 2013) builds upon the NSGA2 algorithm but with a modified survival selection mechanism. To maintain diversity among population members, NSGA3 utilizes well-spread reference points that are constantly supplied and adaptively updated (Gu, Xu, & Li, 2022). The survival selection mechanism (Deb & Jain, 2013) is given in Alg. 2. The use of reference points and NSGA3's efficient niching methodology increases diversity preservation of obtained solutions.

Algorithm 2 Survival selection of NSGA3 Algorithm (Deb & Jain, 2013)

Data: Merged Population $R^{(t)}$, Number of surviving individuals N, Reference Directions Z, Ideal Point Estimation $\bar{z}^*$, Nadir Point Estimation $\bar{z}^{nad}$
Result: Surviving Individuals P^{t+1}
1: $(F_1, F_2, ...) \leftarrow$ non_dominated_sort($R^{(t)}$)
2: S = ∅, $i = 1$
3: **while** $|S| + |F_i| < N$ **do**
4: $S \leftarrow S \cup F_i$; $i = i + 1$
5: **end while**
6: **if** $|S| + |F_L| = N$ **then**
7: $S \leftarrow S \cup F_L$
8: **else**
9: {/* Normalize objectives space and update boundary estimation */}
10: $\bar{S}, \bar{F}_L, \bar{z}^*, \bar{z}^{nad} \leftarrow normalize(S, F_L, \bar{z}^*, \bar{z}^{nad})$
11: {/* niche count, assigned Z_i, perpendicular dist to Z_i */}
12: $\rho, \pi, d \leftarrow 0$
13: **for** $k \leftarrow 1$ to $|S|$ **do**
14: $\pi_k d_k \leftarrow associate(\bar{S}_k, Z)$; $\rho_{\pi_k} \leftarrow \rho_{\pi_k} + 1$
15: **end for**
16: {/* Remaining individuals from F_L to fill up S */}
17: $S \leftarrow S \cup niching(\bar{F}_L, n - |S|, \rho, \pi, d)$
18: **end if**
19: $P^{(t+1)} \leftarrow S$
20: **return** $P^{(t+1)}$

2.5.3. SMSEMOA

Beume proposed SMSEMOA (Beume, Naujoks, & Emmerich, 2007), which aims to maximize the dominated hypervolume in the optimization process by covering a maximal hypervolume with a finite number of points. The algorithm features a selection operator based on the hypervolume measure and non-dominated sorting, resulting in a well-distributed set of solutions that focus on interesting regions of the Pareto front. The hypervolume indicator, also known as the S-Metric, is a commonly used measure for evolutionary multi-objective algorithms (Zitzler, Thiele, Laumanns, Fonseca, & Da Fonseca, 2003).

Alg. 3 describe the fundamental process. To start, a population P_0 consisting of μ individuals is randomly initialized. Then, for each iteration t, SMSEMOA selects parents randomly and applies crossover and mutation operators to generate a new offspring $q_t + 1$. The new offspring is added to the population, resulting in a new population Q with $\mu+1$ individuals. To maintain the population size of μ, a REDUCE method is applied. This cycle continues until the stop criterion is met.

Alg. 4 presented the pseudo code for the reduce method. Initially, the merged population Q is divided into v layers, $\mathcal{R}_1, \ldots, \mathcal{R}_v$, using fast nondominated sorting. Here, R_1 represents the subset of non-dominated points, followed by R_2, which represents the set of non-dominated solutions in $Q \backslash R_1$, and so on. The least ranked front is R_v, and an individual is removed from this front in the selection step to reduce the population size from $\mu + 1$ to μ.

Algorithm 3 The basic steps of the SMSEMOA.

Data: μ the population size
Result: P_{t+1}
1: $P_0 \leftarrow init()$
2: $t \leftarrow 0$
3: **repeat**
4: $q_{t+1} \leftarrow generate(P_t)$ /* generate one offspring by variation operators */
5: $Q \leftarrow P_t \cup \{q_{t+1}\}$
6: $P_{t+1} \leftarrow Reduce(Q)$
7: $t \leftarrow t + 1$
8: **until** A termination criteria is satisfied

Algorithm 4 The basic steps of REDUCE method.

1: $\{\mathcal{R}_1, \ldots, \mathcal{R}_v\} \leftarrow$ non_dominated_sort(Q) /* all v fronts of Q */
2: $r \leftarrow \text{argmin}_{s \in \mathcal{R}_v}[\Delta\phi(s, \mathcal{R}_v)]$
3: $Q \leftarrow (Q \backslash \{r\})$

2.5.4. AGEMOEA

The AGEMOEA is an evolutionary algorithm proposed by Panichella (2019). This algorithm predicts the shape of the Pareto front and uses this estimate to guide selection and diversity. The basic steps of the algorithm are given in Alg. 5. AGEMOEA is based on NSGA2 but replaces crowding distance with a survival score for non-dominated fronts that combines diversity and proximity. AGEMOEA uses the L_p norm instead of the L_2 norm for accurate measurement in M-dimensional space, which differs from the NSGA-III algorithm (Shen & Pan, 2023). The Euclidean norm (L_2 norm) is a specific type of L_p norm where p is equal to 2. The value of p determines the distance measurement in $\mathbb{R}^m$. In AGEMOEA, Eq. (1) can be used to calculate the value of p (Line 6), where C_i represents the ith coordinate of C in the objective space. Last, Eq. (2) can be used to calculate the survival score, which strongly depends on diversity and proximity (Line 9).

Algorithm 5 Basic steps of AGEMOEA (Panichella, 2019)

Data: Number of objectives M, Population size N
Result: Final population P
1: $P \leftarrow$ RANDOM-POPULATION(N)
2: **while** not(stop_condition) **do**
3: $Q \leftarrow$ GENERATE-OFFSPRING(P)
4: $\mathbb{F} \leftarrow$ FAST-NONDOMINATED-SORT($P \cup Q$)
5: $\mathbb{F} \leftarrow$ NORMALIZE($\mathbb{F}$)
6: $p \leftarrow$ GET-GEOMETRY($\mathbb{F}_1, M$)
7: $d \leftarrow 1$ //First non-dominated rank
8: **while** $|P| + |\mathbb{F}_d| \leq N$ **do**
9: SURVIVAL-SCORE($\mathbb{F}_d, d, p$)
10: $P \leftarrow P \cup \mathbb{F}_d$
11: $d \leftarrow d + 1$
12: **end while**
13: SORT($\mathbb{F}_d$) //by survival scores
14: $P \leftarrow P \cup \mathbb{F}_d[1 : (N - |P|)]$
15: **end while**
16: **return** P

$$p = -\frac{\log(M)}{\log\left(\frac{1}{M}\sum_{i=1}^{M} C_i\right)} = \frac{\log(M)}{\log(M) - \log\left(\sum_{i=1}^{M} C_i\right)} \quad (1)$$

$$score(S) = \frac{diversity(S, \mathbb{F}_1)}{proximity(S)} \quad (2)$$

2.6. Fitness functions

For this research, six distinct fitness functions were designed to represent diverse attributes that ought to be present in a menu. Each fitness function was evaluated separately from the rest. These functions were given below. For all fitness values, a smaller value indicated better fitness.

- Color
- Consistency
- Main Ingredients
- Nutrient
- Repetition
- Meal Group

The details of these fitness functions were described in the following sections.

2.6.1. Color

$$fitness_{color} = 1 - \frac{\sum_{d=1}^{n} \frac{unique(colors_d)}{\#\text{ of dish types}}}{n} \tag{3}$$

2.6.2. Consistency

The calculation of this fitness function is based on the balance of solid and liquid dishes on the menu. It is important to avoid having all solid or all liquid dishes on a day's menu. Ideally, there should be a balance between the two, with a preference for more solids than liquids. The equation for calculating this fitness function can be found in Eqs. (4) and (5). While n indicates the maximum number of days, d indicates the current day.

$$f(d)\begin{cases} 0, & \text{2 solid and 1 liquid} \\ 1, & \text{3 liquid/solid} \\ 0.5, & \text{1 solid and 2 liquid} \end{cases} \tag{4}$$

$$fitness_{consistency} = \frac{\sum_{n}^{d=1} f(d)}{n} \tag{5}$$

2.6.3. Main ingredients

The calculation of this fitness function is based on the variation of the main ingredients used in the menu. It takes into account the essential components of the dishes served on two consecutive days. The first day's calculation involves determining the average number of unique ingredients, while the values of the subsequent days are calculated in conjunction with those of the prior day. The final fitness value of the main ingredients is derived from the average of all the calculated values. The calculation of this function is given in Eqs. (6) and (7). n indicates the maximum number of days, and d indicates the current day.

$$f(d) = \begin{cases} \frac{unique(\text{main ingredients}_d)}{\#\text{ of dish types}}, & \text{d=1} \\ \frac{unique(\text{main ingredients}_{d-1} + \text{main ingredients}_d)}{\#\text{ of dish types} * 2}, & \text{otherwise} \end{cases} \tag{6}$$

$$fitness_{\text{main ingredient}} = 1 - \frac{\sum_{\text{d=1}}^{n-1} f(d)}{n-1} \tag{7}$$

2.6.4. Nutrient

The nutrient fitness function is calculated based on the menu's energy, carbohydrate, protein, and fat content. According to nutritional guidelines, these values are not determined by a single absolute value. Rather, values within a certain tolerance factor (λ) are considered appropriate. This fitness function takes this into consideration. This equation is given in Eqs. (8) and (9). Here, target values are treated as variables to enable menu planning in special groups. The values in our study are given in the experimental setup section.

$$f(\text{nutrient}_i) = \begin{cases} 1, & (1-\lambda) * \text{target}_i \le \text{nutrient}_i \le (1-\lambda) * \text{target}_i \\ 1 - \frac{|\text{target}_i - \text{nutrient}_i|}{\text{target}}, & otherwise \end{cases} \tag{8}$$

$$1 - \frac{\sum_{i=1}^{4} f(\text{nutrient}_i)}{4} \tag{9}$$

2.6.5. Repetition

The repetition fitness function (Eq. (10)) is calculated based on the diversity of dishes over n days. In this equation, d indicates the current day.

$$1 - \frac{\text{unique}(\sum_{d=1}^{n} \text{dishes}_d)}{\text{day} * \#\text{ of dish types}} \tag{10}$$

2.6.6. Meal group

To provide a comprehensive overview of the dishes, they have been categorized into 19 groups (details are given in Table 1). The meal group fitness (Eq. (11)) is determined by the frequency of these groups appearing on two consecutive days. n and d indicate the maximum number of days and current day, respectively.

$$1 - \frac{\sum_{d=1}^{n-1} \frac{unique(\text{meal group}_d + \text{meal group}_{d+1})}{\#\text{ of dish types} * 2}}{\text{n - 1}} \tag{11}$$

2.7. Evaluation of the menu samples by dietitians

In our study, the best and worst menus produced by four algorithms were presented to food service dietitians to evaluate their practical applicability. An announcement was made via social media that food service dietitians are needed to evaluate generated menus. The study included 20 dietitians who had been working in food services for at least five years and volunteered to participate. Five of the dietitians had a doctorate (25%), 8 had a master's degree (40%), and all were engaged in menu planning.

First, from the menus generated by the algorithms, the best and worst performing samples (8 in total) were selected according to the average fitness value in all runs. Then, a menu evaluation form (Given as Supplementary Material in Table S1) containing various criteria was created for the food service dietitians to evaluate generated menus. Menus and menu evaluation forms were sent to food service dietitians as an online questionnaire. Both the researcher who sent the questionnaire and the dietitians who administered the questionnaire did not know which menu was produced by which algorithm (double-blind study design). The questionnaire consisted of 8 questions. Dietitians rated the menus between 0 (not appropriate) and 5 (very appropriate) scores according to these criteria. The suitability of the menus was assessed based on the scores provided. Furthermore, dietitians provided feedback regarding areas that required improvement.

2.8. Statistical analysis

Data analysis was carried out using the R language (Version 3.6.1). The mean ± standard deviation values were provided, while the median values were enclosed in parentheses. The best results for each metric are highlighted in bold. To ensure the reliability of the results, each experiment was repeated 30 times with different random seeds, and the Kruskal–Wallis test was utilized for analysis. If there was a significant difference between the groups, the Mann–Whitney U test was used for pairwise comparisons. When parameters with different superscript letters (a,b) are statistically significantly different from each other. A p-value < 0.05 was considered statistically significant for all analyses and given in bold.

3. Experimental study

3.1. Experimental setup

The experiments were carried out on a platform with an Intel i5 12400 2.50/4.40 GHz processor and 32 GB of RAM. While the EvoMeal was created using Python 3.10, Pymoo[3] was used for implementing all

[3] https://www.pymoo.org.

Table 2
Evaluation of the effectiveness of algorithms based on their fitness function values.

Metrics	AGEMOEA	SMSEMOA	NSGA2	NSGA3	*p*-value
Color	0.0000 ± 0.0000^{a}	0.0044 ± 0.0243^{a}	0.0400 ± 0.0603^{b}	0.0267 ± 0.0365^{b}	**0.0001**
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	
Consistency	0.0000 ± 0.0000^{a}	0.0000 ± 0.0000^{a}	0.0267 ± 0.0740^{b}	0.0033 ± 0.0183^{ab}	**0.0290**
	(0.0000)	**(0.0000)**	(0.0000)	(0.0000)	
Main ingredients	0.0569 ± 0.0344^{a}	0.0351 ± 0.0472^{b}	0.1569 ± 0.1502^{ac}	0.1652 ± 0.0707^{c}	**0.0000**
	(0.0571)	**(0.0000)**	(0.1131)	(0.1685)	
Nutrient	0.0251 ± 0.0336^{a}	0.0282 ± 0.0385^{a}	0.0648 ± 0.0754^{ab}	0.0669 ± 0.0416^{b}	**0.0005**
	(0.0182)	(0.0198)	(0.0419)	(0.0631)	
Repetition	0.0022 ± 0.0122^{a}	0.0267 ± 0.0450^{b}	0.0711 ± 0.1262^{b}	0.0467 ± 0.0558^{b}	**0.0008**
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	
Meal group	0.0042 ± 0.0127^{a}	0.0056 ± 0.0144^{a}	0.0278 ± 0.0442^{b}	0.0514 ± 0.0565^{b}	**0.0000**
	(0.0000)	(0.0000)	(0.0000)	(0.0417)	
Average	0.0299 ± 0.0206^{a}	0.0357 ± 0.0242^{b}	0.0806 ± 0.0277^{c}	0.0691 ± 0.0229^{d}	**0.0000**
	(0.0249)	(0.0294)	(0.0790)	(0.0694)	

the algorithms. Additionally, statistical tests were conducted using the R language.

The total number of evaluations of the algorithms was chosen as 10 000. The population size was 20. Other control parameters were left as the default value set by Pymoo.

In the nutrient fitness function (Eq. (8)), the $target_{energy}$ is 720, the $target_{protein}$ is 28, the $target_{carbohydrate}$ value is 85, and the $target_{fat}$ is 26. The chosen values align with the energy and nutrient needs of moderately/actively physically active adults, as specified in the TUBER. To account for a single meal, we have considered 2/5 of the requirement, and the value of one bread roll has been included in each menu, resulting in the adjustment of these values. Finally, the tolerance factor (λ) in Eq. (8), which was used to calculate these values, was chosen as 0.33.

3.2. Comparison of the algorithms

Evaluation of the effectiveness of algorithms based on their fitness function values was given in Table 2. According to the descriptive statistics results (median>mean>standard deviation), while AGEMOEA achieved the best result in all fitness functions except the main ingredients, SMSEMOA was the most successful algorithm in the main ingredients fitness function. In terms of consistency, AGEMOEA and SMSEMOA showed the same performance. NSGA2 and NSGA3 achieved similar performance in terms of all fitness functions.

According to the statistical analysis, AGEMOEA and SMSEMOA algorithms were found to be the best performers among the four algorithms in terms of color and meal group fitness functions ($p < 0.05$). In the consistency fitness function, NSGA3 achieved significantly better results along with AGEMOEA and SMSEMOA while in the nutrient fitness function, NSGA2 performed significantly better alongside AGEMOEA and SMSEMOA ($p < 0.05$). SMSEMOA was the most successful algorithm in main ingredients, while AGEMOEA was the most successful algorithm in repetition ($p < 0.05$). In addition, based on the average fitness values, the algorithms were ranked as AGEMOEA, SMSEMOA, NSGA3, and NSGA2, respectively ($p < 0.001$) (Table 2).

The comparison of the energy and nutritional contents of the menus generated by the algorithms was summarized in Table 3. It has been observed that all algorithms provided energy, carbohydrate, protein, and fat between the reference values in the menus, and there was no statistically significant difference between them ($p > 0.05$).

Expressing the stochastic behavior of an entire population through a simple function of population size or evaluation number is difficult due to complex interactions between individuals, search operators, and control parameters (Lehre & Yao, 2009). Hence, Table 4 and Fig. 4 presented an experimental time analysis to evaluate the time complexity of the algorithms. These tables and figures also showed Hypervolume, generational distance (GD), and Inverted Generational Distance (IGD) metrics. Hypervolume calculation, which is used to evaluate the performance of multi-objective optimization algorithms, determines the area dominated by a particular set of solutions, relative to a reference point. Reference points were selected as 1.2 for each fitness value. The General Distance (GD) Performance Indicator quantifies the distance from a solution to the Pareto front. It provides insights into how well a solution approximates the optimal trade-offs between conflicting objectives. A smaller GD value indicates better proximity to the true Pareto front. On the other hand, IGD takes a different approach. Instead of measuring the solution's distance to the Pareto front, it evaluates the distance of the reference solution set to the solution set generated by an optimization algorithm. Upon examining the Table 4, while AGEMOEA and SMSEMOA algorithms were similar in Hypervolume, GD, and IGD metrics, these two algorithms produced better results than NSGA2 and NSGA3. When we compared the NSGA2 and NSGA3, NSGA2 produced better results than NSGA3 regarding the hypervolume metric. However, it generated worse results in terms of GD and IGD metrics than NSGA3. To put it simply, the NSGA2 covered a larger Pareto front than the NSGA3. Since it fell behind in GD and IGD metrics, it means that the solution set of the NSGA2 was further away from the actual Pareto fronts. In other words, the NSGA3 achieved better convergence than the NSGA2. After examining the running time of the algorithms, it was evident that the NSGA2 was the fastest, followed by the AGEMOEA and SMSEMOA, respectively. However, the calculation time of the NSGA3 was higher than the other three algorithms. When analyzing the boxplot for the Hypervolume metric, the SMSEMOA and AGEMOEA performed exceptionally well in most runs with minimal standard deviation. While the plots for NSGA2 and NSGA3 appeared comparable, NSGA2 showed slightly better results regarding the Hypervolume metric. When GD and IGD metrics were examined, SMSEMOA and AGEMOEA achieved the best results. Unlike the Hypervolume metric, NSGA3 appeared better than NSGA2 with respect to GD and IGD metrics. This meant the NSGA2 algorithm generated solutions with lower robustness. Regarding calculation time, the algorithms' standard deviation values appeared relatively low. However, while NSGA2 was the fastest algorithm, NSGA3 tended to run quite slowly.

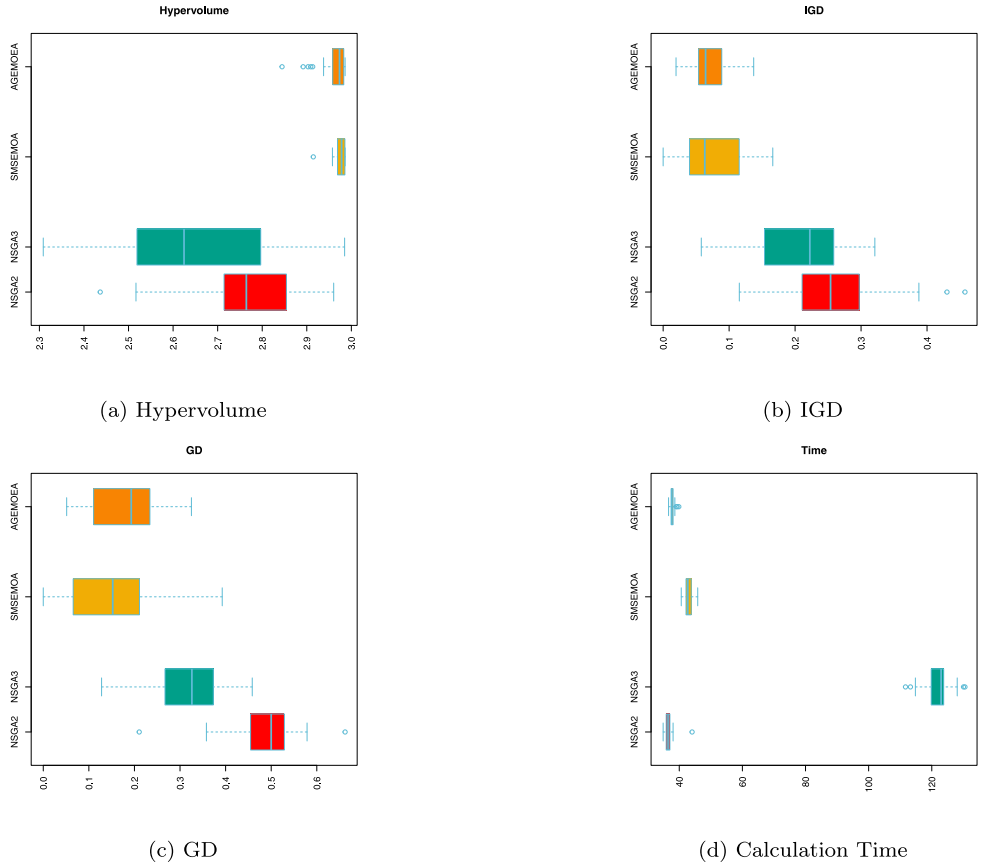
Finally, Fig. 5 displayed the convergence plots for the algorithms. Upon examining the convergence plot for Hypervolume, NSGA2, and NSGA3 initially appeared comparable. However, as the iterations progressed, NSGA2 outperformed NSGA3, achieving better results. In contrast, the AGEMOEA and SMSEMOA algorithms showed a superior starting value compared to other algorithms. Additionally, during the first half of the search, AGEMOEA and SMSEMOA yielded better outcomes than other algorithms' end results, with a slight improvement in the second half. The algorithms' behavior in GD and IGD metrics is quite similar. In both metrics, the NSGA2 algorithm's initial values were the worst. Afterward, it was able to make very minimal improvement. This showed that the algorithm had problems with convergence. Although the NSGA3 algorithm converged better, it fell behind the AGEMOEA and SMSEMOA algorithms, which showed continuous improvement throughout the search. When examining the overall fitness values, it appeared that NSGA2 and NSGA3 completed their searches

Table 3
Comparison of the energy and nutrient contents of the menus generated by the algorithms.

Metrics	AGEMOEA	SMSEMOA	NSGA2	NSGA3	*p*-value
Energy	677.4339 ± 32.5370 (677.6390)	661.9723 ± 42.2287 (657.3760)	670.0608 ± 52.9758 (669.6910)	674.4539 ± 43.1818 (665.9850)	0.4679
Cho	73.1090 ± 6.9872 (72.9110)	69.9317 ± 6.0768 (68.6480)	70.7861 ± 10.9755 (73.3740)	72.3399 ± 8.0600 (71.2850)	0.3393
Protein	31.8569 ± 2.1595 (31.8740)	31.3695 ± 2.6027 (31.7570)	32.0779 ± 2.9646 (32.3000)	31.0720 ± 2.9090 (30.9880)	0.4383
Fat	27.6561 ± 1.7132 (27.2730)	27.4770 ± 2.1638 (27.2450)	27.6820 ± 2.5374 (27.3740)	27.8419 ± 2.5077 (27.7320)	0.9528

Table 4
Evaluation of the algorithms in terms of Hypervolume, IGD, GD, and calculation time.

Metrics	AGEMOEA	SMSEMOA	NSGA2	NSGA3	*p*-value
Time	37.8010 ± 0.7175^{a} (37.6714)	42.9060 ± 1.2964^{b} (42.7214)	36.6626 ± 1.5913^{c} **(36.4417)**	122.0578 ± 4.3989^{d} (122.9564)	**0.0000**
Hypervolume	2.9610 ± 0.0346^{a} (2.9735)	2.9746 ± 0.0143^{a} **(2.9781)**	2.7559 ± 0.1323^{b} (2.7644)	2.6420 ± 0.1780^{c} (2.6245)	**0.0000**
IGD	0.0680 ± 0.0290^{a} (0.0643)	0.0772 ± 0.0472^{a} **(0.0631)**	0.2572 ± 0.0837^{b} (0.2538)	0.2072 ± 0.0698^{c} (0.2224)	**0.0000**
GD	0.1803 ± 0.0790^{a} (0.1930)	0.1533 ± 0.0960^{a} **(0.1524)**	0.4879 ± 0.0816^{b} (0.5000)	0.3173 ± 0.0804^{c} (0.3260)	**0.0000**



(a) Hypervolume (b) IGD

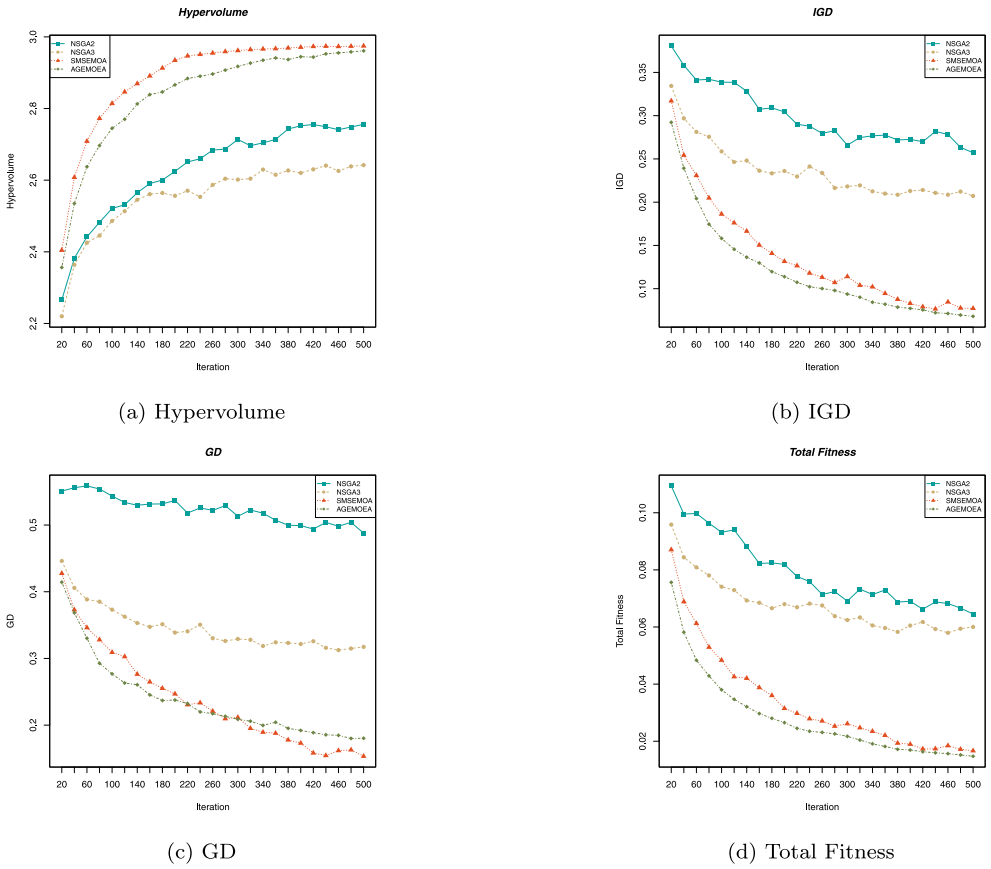
(c) GD (d) Calculation Time

Fig. 4. Boxplot analysis of the algorithms in terms of hypervolume, IGD, GD, and calculation time.

at comparable levels despite NSGA2 having a weaker starting fitness value. Conversely, AGEMOEA displayed better initial values than the other algorithms and improved consistently throughout the search process. Although SMSEMOA was initially worse than AGEMOEA, the search eventually produced a similar result to AGEMOEA.

3.3. Evaluation of the best and worst menu samples by dietitians

The study's algorithms are randomness-based. Therefore, to ensure the reliability of the results, each experiment was repeated 30 times with different random seeds. In these studies, dietitians evaluated the best and worst results generated by the algorithms. The best and worst menus generated by algorithms were given Table 5. Evaluation of the menu samples by dietitians was summarized in Table 6 and Fig. 6. According to all criteria, except consistency, the best menus of AGEMOEA and SMSEMOA received similar scores and had a significantly higher average score than those of the other two algorithms ($p < 0.001$). In terms of consistency, the scores of the best menus of the AGEMOEA, SMSEMOA, and NSGA2 algorithms were similar to each other and higher than those of NSGA3 ($p = 0.009$). According to the "color" and



(a) Hypervolume (b) IGD

(c) GD (d) Total Fitness

Fig. 5. A graphical plot of the performances of the different search algorithms.

"taste" scores, the worst menus of AGEMOEA, SMSEMOA, and NSGA2 were similar ($p > 0.05$), while the worst menus of NSGA3 had the lowest score significantly ($p < 0.05$). According to other criteria, the scores of the worst menus of all algorithms were similar ($p > 0.05$).

4. Threats to validity

Our empirical study is based on a tool prototype, which poses threats to internal validity. If there are any faults in the tool, the validity of our conclusions may be compromised. While we have carefully tested EvoMeal, we cannot guarantee that it is entirely bug-free. However, we have made our tool and case study with the dataset available online at https://github.com/omursahin/EvoMeal to mitigate the risk and enable independent replication of our experiments. As our techniques are based on randomized algorithms, it is possible that randomness may affect the results. To address this issue, we repeated each experiment 30 times with different random seeds and used appropriate statistical tests to analyze the results. External validity threats arise from the fact that we only created a menu for one meal for five days with a single dataset. This is due to the difficulty of creating a dataset with too many dishes. We did not compare our tool with others as we are unaware of existing tools that can generate meals.

5. Discussion

Menu planning is a time-consuming problem for dietitians. It is difficult to ensure the nutritional requirements of the target group and to plan a menu completely per the basic menu planning principles. In addition, this problem is a complex optimization problem of the NP-Hard type. For example, 332 meals and 10^6 different menus can be produced in the data set we created. In addition, considering many constraints, it is much more efficient to solve this problem with meta-heuristic algorithms. There are few studies on this area in the literature (Hernandez-Ocana et al., 2018; Marrero et al., 2020; Moreira et al., 2017; Seljak, 2006, 2009; Stern et al., 2023). Existing studies have limited constraints, and the fact that they are not open-source makes it difficult to conduct more comprehensive new studies. In addition, the lack of dietitian opinions limits its applicability in practice. Therefore, in this study, four different algorithms were applied to solve the menu planning problem, and the results were shared with 20 food service dietitians, who were experts in the area, and they were asked to score the menus. In addition, an open-source tool called EvoMeal has been developed. Moreover, we made our generated data publicly available at https://doi.org/10.5281/zenodo.10845859.

The AGEMOEA and SMSEMOA algorithms outperformed other algorithms in solving the fitness functions we have determined, which makes them more successful in expert evaluation. One of the main reasons for the superiority of the SMSEMOA is that it uses the hypervolume indicator in its selection mechanism. Compared to the more complex sorting and crowding distance calculations in NSGA2 and the reference point system in NSGA3, SMSEMOA's focus on hypervolume may have made it conceptually simpler and easier to implement, while also providing more robust performance. However, SMSEMOA also has some disadvantages. The main drawback of SMSEMOA is its computational expense, especially as the number of objectives increases (Shang & Ishibuchi, 2020). Calculating the hypervolume is significantly more computationally intensive than the mechanisms used in NSGA2 and NSGA3, making SMSEMOA less practical for problems with a large number of objectives or when fast execution times are required. Due to its computational complexity, SMSEMOA may not scale as well as AGEMOEA or NSGA3 to huge problem sizes or extremely high-dimensional objectives. AGEMOEA employs a geodesic distance-based method that can better represent the shape of the Pareto front and

Table 5
The best and worst menus generated by algorithms.

	#	AGEMOEA	SMSEMOA	NSGA2	NSGA3
Best	1	Peach Kebab Corn Rice Pilaf Watermelon	Stuffed Peppers with Meat Green Beans with Olive Oil Black Cabbage-Carrot Salad	Meatball with Egg Yolk & Lemon Sauce Pastry with Potato Apricot	Stuffed Peppers with Meat Leeks with Olive Oil Nectarine
	2	Bean soup Spinach with egg Pudding	Tutmac Soup Simit Kebab Quince Dessert	Bean Soup Steamed Whiting Shepherd's Salad	Tutmac Soup Bitlis Meatballs Eggplant Salad with Tomatoes
	3	Juicy Meatballs Rice Pilaf Kiwi	Spinach with egg Potato Roll Pastry Bostana Salad	Turnip Moussaka Walnut Noodles Banana	Turnip Moussaka Pastry with Potato Melon
	4	Green Lentil Soup Zucchini with minced meat (with yoghurt) Lentil Salad	Toyga Soup Şeftali Kebab Grape Compote	Lentil soup Grilled Mackerel Lentil Salad	Lentil Soup Spinach with Egg Apple Dessert with Cream
	5	Mackerel Pasta with Napolitan Sauce Mixed Fruit Hosaf	Turnip Moussaka Rice Pilaf Sour Grape Salad	Spinach with egg Rice Pilaf Dried Apricot Compote	Peach Kebab Corn Rice Pilaf Fig
Worst	1	Tas Kebab with Quince Bulgur Pilaf with Vegetables Kefir	Tutmac Soup Meat Roast with Potato Puree Eggplant Salad with Tomatoes	Stuffed Peppers with Meat Pasta with tomato sauce Zerde	Meatball with Egg Yolk & Lemon Sauce Green Beans with Olive Oil Gavurdagi Salad
	2	Red Lentil Soup Mitit Meatballs with Puree Eggplant Salad with Tahini	Spinach with egg Pasta with Napolitan Sauce Apricot	Turnip Moussaka Pastry with Cheese Peach	Spinach with Egg Rice Pilaf Kazandibi Dessert
	3	Simit Kebab Fried Cauliflower with Yogurt Grape	Bean Soup Simit Kebab Milk pudding	Stuffed Tomatoes with Meat Bulgur Pilaf with Chickpeas Sour Grape Salad	Toyga Soup Grilled Mackerel Dried Plum Compote
	4	Chickpeas with Meat Rice Pilaf Purslane Salad	Turnip Moussaka Pastry with Potato Mandarin	Baked Chicken Zucchini Patty with Olive Oil Peach Compote	Yayla Soup Zucchini Moussaka Pear Dessert
	5	Yayla soup Turnip Moussaka Quince Dessert	Juicy Meatballs Dried Black-Eyed Peas with Olive Oil Chicken Breast Pudding	Lentil Soup Zucchini with minced meat Cucumber-Yoghurt	Meatball with Egg Yolk & Lemon Sauce Pasta with Tomato Sauce Melon

Table 6
Evaluation of the best and worst menus samples by dietitians.

Criteria	Best					
	AGEMOEA	SMSEMOA	NSGA2	NSGA3	F	*p*-value
Taste	3.95 ± 0.76^{a}	**4.00 ± 0.46**^{a}	3.00 ± 0.46^{b}	3.15 ± 0.49^{b}	**17.773**	**<0.001**
Color	**3.95 ± 0.89**^{a}	3.80 ± 0.77^{a}	3.10 ± 0.31^{b}	3.20 ± 0.41^{b}	**8.814**	**<0.001**
Consistency	**4.00 ± 0.73**^{a}	3.90 ± 0.72^{ab}	3.30 ± 1.03^{ab}	3.30 ± 0.80^{b}	**4.149**	**0.009**
General meal pattern	3.90 ± 0.85^{a}	**3.90 ± 0.72**^{a}	3.10 ± 0.96^{b}	3.15 ± 0.37^{b}	**6.939**	**<0.001**
Dist. of main dishes	4.00 ± 1.03^{a}	**4.20 ± 0.77**^{a}	3.10 ± 0.72^{b}	2.90 ± 0.55^{b}	**13.533**	**<0.001**
Dist. of side dishes	3.90 ± 0.97^{a}	**4.20 ± 0.41**^{a}	3.20 ± 0.83^{b}	3.20 ± 0.77^{b}	**8.565**	**<0.001**
Dist. of fruit/dessert/salad or dairy products	4.15 ± 0.75^{a}	**4.30 ± 0.66**^{a}	3.30 ± 0.80^{b}	2.90 ± 1.17^{b}	**12.114**	**<0.001**
General variety of meal	**4.05 ± 0.89**^{a}	4.00 ± 0.65^{a}	3.00 ± 0.65^{b}	2.30 ± 0.80^{b}	**25.091**	**<0.001**
Overall score	3.97 ± 0.66^{a}	**4.04 ± 0.49**^{a}	3.14 ± 0.57^{b}	3.03 ± 0.45^{b}	**18.806**	**<0.001**
Criteria	Worst					
	AGEMOEA	SMSEMOA	NSGA2	NSGA3	F	p-value
Taste	2.90 ± 0.55^{a}	**3.10 ± 0.55**^{a}	2.75 ± 0.44^{a}	2.00 ± 0.79^{b}	**12.817**	**<0.001**
Color	2.95 ± 0.89^{a}	**3.10 ± 0.91**^{a}	2.90 ± 0.55^{a}	2.20 ± 0.77^{b}	**5.113**	**0.003**
Consistency	2.80 ± 0.52	2.85 ± 0.37	2.80 ± 0.62	2.50 ± 0.95	1.219	0.309
General meal pattern	3.05 ± 0.89	2.90 ± 0.55	2.90 ± 0.55	2.80 ± 0.77	0.428	0.734
Dist. of main dishes	3.05 ± 0.76	2.90 ± 0.72	2.90 ± 0.55	2.80 ± 0.62	0.479	0.698
Dist. of side dishes	2.70 ± 0.86	2.80 ± 0.89	2.90 ± 1.25	2.90 ± 0.85	0.191	0.902
Dist. of fruit/dessert/salad or dairy products	2.80 ± 0.70	2.90 ± 0.72	2.80 ± 1.29	2.30 ± 1.03	2.064	0.112
General variety of meal	2.65 ± 0.59	2.60 ± 0.50	2.90 ± 0.45	2.60 ± 0.94	0.981	0.406
Overall score	2.99 ± 0.41	2.96 ± 0.43	2.91 ± 0.44	2.65 ± 0.70	0.858	0.598

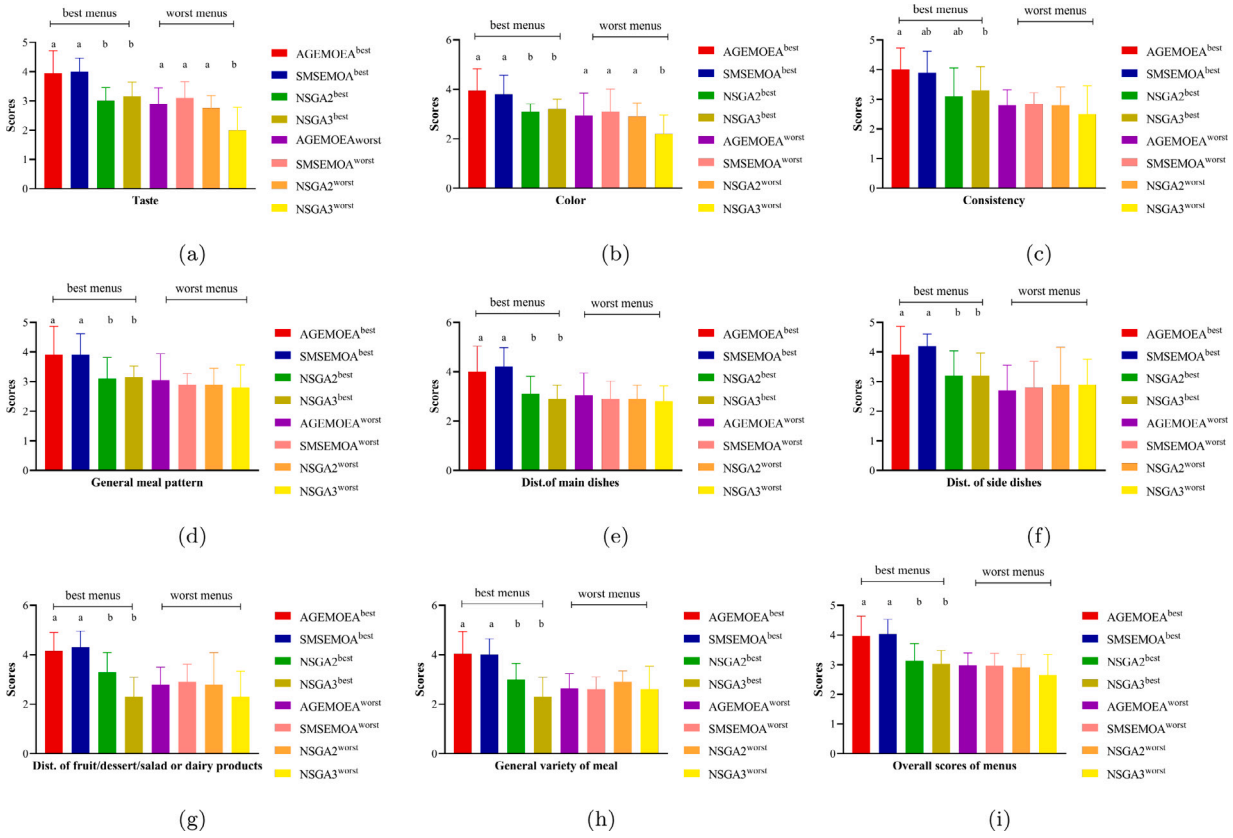


Fig. 6. Evaluation of the menu samples by dietitians.

generate solutions that are well-distributed along it. This makes AGEMOEA a highly promising algorithm. The NSGA2 algorithm is not as effective as other algorithms. It is typically used for solving multi-objective problems that involve three or fewer objectives. However, our proposed problem has six objectives, which makes it more suitable for solving with many-objective algorithms. Consequently, algorithms such as AGEMOEA and SMSEMOA, which are specifically designed for many-objective problems, have generated much more successful results.

The dataset we created within the scope of this study consists of 332 foods from Turkish cuisine. In addition to nutritional values such as energy, protein, and carbohydrate, which are considered important in the menu planning principles of these foods, consistency, dish type, color, main ingredients, and meal group information are also included. Within the scope of the study, six different fitness functions calculated using this information were defined. These functions have been designed in accordance with general menu planning principles. However, different cultures and different cuisines may have different requirements. Considering this, a mechanism has been created that makes it very easy to add new fitness functions. Practitioners simply need to inherit the AbstractFitness class by creating a new class, defining the fitness() method, and then adding it to the fitness function set in the settings. Thus, the algorithm will work by taking this objective into account.

6. Conclusion

In conclusion, AGEMOEA and SMSEMOA have been used for the first time in the literature for the menu planning problem, which is a multi-objective problem, and it has been observed that they perform better than both NSGA2 and NSGA3 in general. In addition, expert opinions have confirmed that AGEMOEA and SMSEMOA generally give better results than other algorithms. Moreover, experts' scores show that the fitness functions we used in the menu planning problem were modeled correctly.

This study has several strengths. First, the application is open-source and has a flexible infrastructure that can be updated to meet the needs of researchers. Secondly, practitioners can easily use the application to solve problems in the field. In addition, using variables enables customization of menus for different health conditions by adjusting target values. Lastly, this study is the first to gather the opinions of experts on algorithm-generated menus.

While the study results may not be significant for all parameters, they indicate that menus created by AGEMOEA and SMSEMOA received the highest score, which is a promising outcome. However, there are aspects of the study that need improvement. This study is mostly focused on tool development. However, future studies need to conduct a more comprehensive expert evaluation to increase its applicability to the field. For example, conducting some sensitivity analysis or robustness checks on the expert scores would be beneficial. Moreover, the tool should be improved with suggestions from the dietitians. New constraints (such as cost, seasonality, and sustainability) should be added to increase expert scores, and the data set should be expanded.

In addition, experts have reported that one problem is obtaining accurate results on "palatal taste" and "local habits". To solve this problem, it is essential to plan large population studies. Another important point for dietitians is the concept of glycemic index. The glycemic index is measured based on the rate at which foods raise blood sugar. Due to this situation, giving high glycemic index foods together is not appropriate. Dietitians who participated in the study emphasized this situation in several menus. Therefore, calculating the glycemic index of the menus and generating a menu with a lower glycemic index will increase the scores. Furthermore, the results obtained by implementing other algorithms in the literature could be evaluated. Finally, the control parameters of the algorithms could be analyzed, and their performance should be investigated.

CRediT authorship contribution statement

Omur Sahin: Conceptualization, Methodology, investigation, visualization, Software, Writing – original draft, Writing – review & editing. **Gizem Aytekin-Sahin:** Conceptualization, Methodology, investigation, visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

While the source code is available on Github, the outputs of the paper can be accessed on Zenodo.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.eswa.2024.124213.

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